

It knows you're there. Here's how to set it up.

A 24 GHz mmWave radar that detects people — even sitting perfectly still — and shows up in Google Home as an occupancy sensor you can use to start automations. Hand-built firmware in Rust on an ESP32-C6; treat it as a friendly prototype.

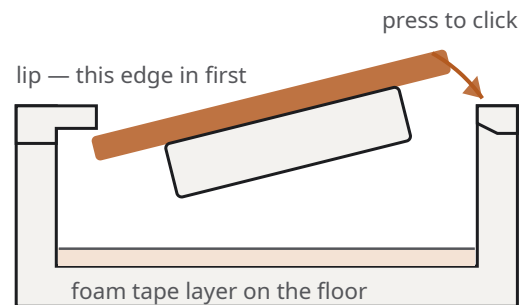
1 Check what you need

- **A Thread border router** on your Wi-Fi: any of Nest Hub (2nd gen), Nest Hub Max, Nest Wifi Pro, or Google TV Streamer. No border router = it can't join.
- **Google Home app** on your phone, with Bluetooth on.
- **USB-C power** near the mounting spot (any phone charger).

Before you pair: this is a development device (test Matter IDs, not certified). Text me the Gmail address you use with Google Home first — I need to add it to my developer project or the app will refuse to commission it.

2 Load the sensor into the mount

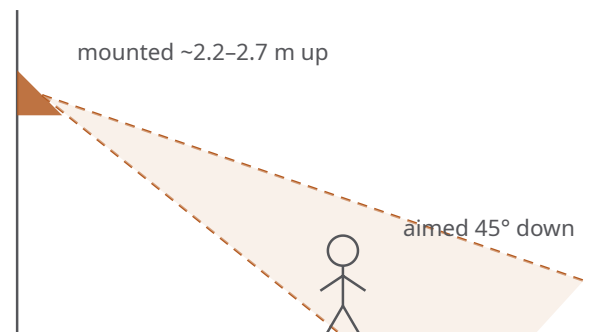
1. Stick **one layer of the foam tape** flat on the pocket floor (the deepest inside face).
2. Hold the sensor **radar side out** (white antenna board faces you), USB-C connector toward the open slot.
3. Slide the board's edge **under the wide lip first**, then press the other edge in until it **clicks past the two small tabs**.
4. To remove: lift the tab edge over the tabs with a fingernail and tilt the board out.



Radar board face out; the tabs hold its edge once it clicks.

3 Pick the spot

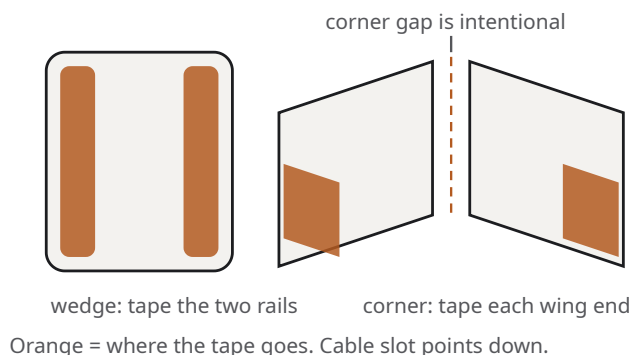
- **Wedge mount:** flat wall, about **2.2-2.7 m up**, aimed into the room. It tilts the radar 45° down.
- **Corner mount:** in the top corner of the room, near the ceiling; it aims diagonally down into the middle of the room.
- The beam is wide ($\pm 60^\circ$) and reaches ~5 m for a walking person, less for someone sitting still — face it toward where people actually are.
- Avoid aiming at fans, big swaying plants, or through thin walls (radar sees through them a little).
- The **cable slot must face the floor** so the cord hangs straight down.



Aim the face at the living area, not along the wall.

4 Stick it to the wall

- Wipe the wall with isopropyl (or at least a dry cloth) and let it dry.
- Foam tape goes **only on the raised strips**: the two vertical rails on the wedge mount's back, or the flat pad at the end of each wing on the corner mount. The rest of the back is deliberately recessed and never touches the wall.
- Press firmly for **30 seconds, right over the taped areas** — on the corner mount press near the wing ends, not the middle.
- The mount weighs ~20 g with the sensor; good foam tape holds it on textured walls and slightly out-of-square corners (the wings flex a little on purpose).



5 Power it & pair with Google Home

1. Route the USB-C cable out of the bottom slot and plug it in. A slow-blinking LED means it's running; a fresh 15-minute pairing window opens at every power-up.
2. In the Google Home app: **Devices** → **Add** → **Matter-enabled device**.
3. When asked for a code, choose *Set up without QR code* and enter:

3497 - 011 - 2332

4. Accept the "device isn't certified" warning (see the note on page 1), pick your Thread network, and give it a room.
5. **Be patient and don't cancel** — slow attempts finish late rather than fail; give it up to 5 minutes before retrying.
6. If a retry is needed: unplug/replug the sensor first (fresh pairing window), and delete any half-added entry from Google Home.

6 Use it

It appears as an **occupancy sensor**. In Google Home automations, use the starter "*Occupancy* → *Occupied* / *Unoccupied*" — lights on when someone's in the room, off a bit after it empties. It flips to Occupied within a second or two of someone entering and clears several seconds after the room is truly empty (radar + a hold time, so it doesn't flicker).

PROBLEM	TRY THIS
Won't pair	Power-cycle the sensor, delete stale entries in Google Home, retry and wait the full 5 minutes. Confirm your Gmail is on the developer project.
Paired but "offline"	Check the border router (Nest Hub etc.) is online; power-cycle the sensor — it rejoins in a few seconds.
Occupied when empty	Fan, curtain, or a big plant in the beam — re-aim it, or move it away from the disturbance.
Start over completely	Pop the sensor out of the mount, hold the tiny BOOT button 3+ seconds while powered: factory reset, then re-pair.